

YFCE Blueprint

A memory-first inference appliance on mature-node silicon: chip, board and filesystem designed as one system, targeting the cheapest hardware that runs a large model at usable speed.

Part I — Blueprint: the machine in two variants — the headless appliance and the **AI PC**, where the model is the operating system and nothing runs Windows or Linux — its modules and interfaces, the decisions proposed, the gates and forks, the phase map, and bills of materials with next-plans for the board, FPGA and test-chip steps.

Part II — Simulation plan: the PC-side campaign S0–S10, sized to the machine the repository lives on.

The plan is detailed through the simulation campaign and stops there. Anything that needs hardware bought is listed, not designed.

Typeset from `docs/blueprint.md` and `docs/simulation.md`; the markdown is canonical.

PART I — BLUEPRINT

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PART II — SIMULATION PLAN

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- S6 — Recalibration of models and docs
- S7 — E2-lite: the software architecture on this PC (G4 at small scale)
- S8 — Prefill and NPU sizing, and the kernel-model choice, on paper
- S9 — The AI OS on this PC: contracts and prototype (G6)
- S10 — Bare-metal proof on this PC: boots into the model, no OS (optional)
- Order, dependencies, exit

Blueprint

The whole project as one plan: what the machine is, how it decomposes, the contracts between its parts, the decisions still open and the order they get closed in. It is detailed **up to and including the simulation campaign**, which is sized to run on the PC this repository lives on. Everything after that — a board, an FPGA, a test chip — is listed as a bill of materials and a next-plan, not designed.

Written 2026-08-16 against the state in `decisions.md`: E1 answered "no at 4-bit", the decode block is currently unjustified, and the honest next move is to prove the system in software. The blueprint takes that seriously rather than working around it — see §6, where the design forks depending on what the representation campaign returns. Revised 2026-08-17 to add the **AI PC** (§3): the same apparatus with human I/O and no operating system — the model is the OS, and the appliance loses Linux with it — and, later the same day, to make it voice-first with a fixed touch/key set and models in charge of apps and settings (`os/`). Revised again after NVIDIA disclosed **RTX Spark**, and then re-footed: §1 no longer treats that as an event but as the first arrival on a commodity curve, which is a different plan. The hardware envelope is assumed to commoditise; **Fork C — the stack on hosts somebody else builds — is the base case**, and silicon is an option this document defers rather than a destination it works toward. **YFCE-M**, the portable variant on socketed LPDDR5X modules, is described the same way: a target envelope, not a committed board.

```
docs/simulation.md    the PC-side campaign, stage by stage (S0-S10)
docs/architecture.md  the design and its invariant — this document assumes it
docs/decisions.md     what is settled; §5 below proposes additions to it
```

1. The machine

One machine whose one job is batch-1 decoding of a large model that stays resident, in three variants that share a runtime, a format and an SoC family. Read them as **target envelopes first and boards second**: each says what the software should assume about memory, and each can be met either by hardware somebody else sells (the base case, §6 Fork C) or by building it (an option, gated). **YFCE-1** is the headless appliance: mains-powered, socketed DDR5, serves models over the network. **YFCE-P** is the AI PC: the same board with a microphone array, speakers, a glanceable touch display and a camera — keyboard optional — operated by voice, with a model as its operating system (§3; the OS's own directory is `os/`). **YFCE-M** is the portable one: the same design on socketed LPDDR5X modules instead of DIMMs, battery-powered (below). None of the three runs Linux.

	reference	why
Memory	8 × 64-bit DDR5-5600 DIMM, 128 GB (12 × 64-bit / 192 GB is the family's wide end)	width buys bandwidth; sockets buy capacity per dollar
Bandwidth	358 GB/s peak, ~301 GB/s achieved at 0.84	<code>model/system.py</code> , to be measured in S5
SoC	28 nm-class; DDR5 PHY + controller, DMA/descriptor engine, decode block, staging SRAM, NPU, CPU cores; YFCE-P adds display, USB 3, multi-channel audio with an always-on wake/AEC path, secure boot	the PHY is the hard part; nothing else needs a leading node
NPU	~20 TOPS int8 (bom.py: 15 mm ²) for YFCE-1; 60–100 TOPS for YFCE-P (§3, S8)	decode needs ~0.6 TMAC/s; the rest is prefill, which the AI PC makes central

	reference	why
Storage	NVMe, PCIe 4 ×4, coded model archives	flash → DRAM hop is where software compression already pays
Board	~14–16 layers, ~1,700-ball FCBGA, 8 DIMM slots, ~150 W PSU; YFCE-P adds DP/HDMI, USB, mic array, speaker amp, confirm keys and a hardware mute switch, Wi-Fi	<code>model/bom.py</code>
Software	one bare-metal runtime (mechanism only) + residency manager (lmz lineage) + the resident model set; no OS	§3

What it delivers, if the models hold (real bandwidth ÷ bytes per token):

model	YFCE-1, 8ch DDR5-5600 (301 GB/s)	12ch DDR5-5600 (452)	YFCE-M, 4× LPCAMM2-8533 (459)	Strix Halo (215 measured)
8B at Q4_K_M (4.5 GB)	67 tok/s	100	102	48
70B at Q4_K_M (39.4 GB)	7.6	11.5	12	5.5
70B int8 (70 GB)	4.3	6.5	6.6	3.1
120B-A5B MoE at 4.5 bpw (67 GB resident, 2.8 GB active)	107	161	164	76
235B-A22B MoE at 4.5 bpw (132 GB resident)	needs the 192 GB build	37	37 (256 GB build)	does not fit
405B at 2.5 bpw (127 GB — capacity, not speed)	2.4	3.6	3.6	does not fit

All four columns are bandwidth ÷ bytes per token, and every one of them is also roughly what an RTX Spark does at 128 GB — see below.

The machine is *not* a prefill engine, not a training box, and not a phone. Time-to-first-token on long inputs will be visibly worse than a leading-edge SoC and the positioning has to say so; §3 says what the AI PC does about it.

The envelope is a commodity — what that leaves

NVIDIA disclosed **RTX Spark** (N1X) this month: 20 Grace ARM cores and a 6,144-core Blackwell GPU on TSMC 3 nm, up to 128 GB of soldered unified LPDDR5X at **~300 GB/s**, ~1 PFLOP of AI compute, in laptops *and* desktops from ASUS, Dell, HP, Lenovo, Microsoft Surface and MSI this autumn, at **~\$2,899** for flagship systems. DGX Spark, the earlier workstation, now starts at \$4,699 after the memory-price rise.

Peak is what a vendor quotes; **achieved** is peak × 0.84, the derating `model/system.py` applies to everything including its own configurations, because treating peak as achievable is how these comparisons usually go wrong. Only Strix Halo's number is measured.

system	node	bus	memory	peak	achieved	8B Q4	120B-A5B	AI compute	price
RTX Spark N1X	N3	256-bit	128 GB LPDDR5X-9600, soldered	~300–307	~252–258	57	92	~1 PFLOP	~\$2,899
DGX Spark GB10	N3	256-bit	128 GB LPDDR5X, soldered	273	229	51	82	~1 PFLOP	\$4,699

system	node	bus	memory	peak	achieved	8B Q4	120B- A5B	AI compute	price
Strix Halo	N4	256-bit	128 GB LPDDR5X-8000, soldered	256	215 measured	48	77	~50 TOPS	\$1,500–2,600
M4 Max	N3E	512-bit	128 GB LPDDR5X-8533, soldered	546	459	102	164	~38 TOPS NPU	in a \$4k laptop
YFCE-1	28 nm	512-bit	128 GB DDR5-5600, DIMMs	358	301	67	107	~20 TOPS	BOM \$462–1,422
YFCE-M	28 nm / 16 nm?	512-bit	256 GB LPDDR5X-8533, LPCAMM2	546	459	102	164	~20 TOPS	BOM ~\$2,010

The premise implies its own commoditisation, and this document was slow to say so. If bandwidth is bought with pins rather than lithography — and a 3 nm part choosing 256 bits is evidence that it is — then nobody holds a position on bandwidth, *this project included*. Spark is not an event that took the envelope away. It is the first arrival on a curve that was always going to arrive, and the curve is the thing to plan against.

The curve. Strix Halo in 2025: 256-bit, 215 GB/s measured, \$1,500–2,600. DGX Spark: 273 GB/s at \$4,699. RTX Spark now: ~300 GB/s at ~\$2,899, in laptops and desktops, from six OEMs. Behind them, LPCAMM2 is putting wide LPDDR into upgradeable modules across the whole PC industry — Intel has demoed Panther Lake at 9600 MT/s — Apple has shipped 512 bits since 2023, and AMD, Qualcomm and MediaTek all have to answer Spark. The DRAM spike is cyclical and will unwind. **128 GB at 300+ GB/s for \$1,000–1,500 by 2028 is a conservative reading of that list, not a bold one.**

Which closes the silicon question on arithmetic, before any gate does. `model/bom.py` budgets ~40 engineers for 2.5 years. Add the campaign (4–7 months), a test chip (6–9 months), bring-up and one respin: **parts in 2029–2030**. That is a program racing a cost curve which arrives first, and the machine is mostly DRAM either way, so the mature-node die saving — \$5 against maybe \$150 — shrinks to a rounding error exactly as the envelope commoditises. The chip does not lose to Spark. It loses to the calendar. This is a fourth reason, independent of G1, G2 and G5, and it is the cleanest of the four.

What is perishable, and what is not. The distinction the rest of this document now hangs on:

perishable	bandwidth	anyone can buy pins; that is the premise
	capacity	density and module counts follow, one generation behind
	the mature-node die saving	a rounding error on a machine that is mostly DRAM, and it shrinks as hosts get cheaper
	width past 256 bits, capacity past 128 GB	the same commodity, one generation later — these were offered here as durable advantages two revisions ago, and they are not
durable	residency — one representation from flash to DRAM to the MAC units, page-granular, no conversion anywhere	a property of software, so it runs on anyone's hardware, and it gets <i>cheaper to deliver</i> as hosts get cheaper
	the operating system (os/)	no hardware roadmap produces it

Read the roofline again with that in mind. `tokens/s = bandwidth ÷ bytes per token`. Commoditisation hands the numerator to everyone. The denominator is the half this project owns — and E1 measured its headroom at 4.5% against `Q4_K_M`, which is not a moat either. So what the format is worth is **not ratio, it is residency**: cold start, model and expert swap, resident footprint, and no conversion between the bytes on flash and the bytes the matrix units consume. That is G4, measured in S7, and it moves from a supporting experiment to the load-bearing one.

Cheap hosts are the enabling condition, not the threat. Every \$1,200 128 GB box that ships is a machine this stack can run on and did not have to build, fund or sell. A software layer that needs cheap wide memory to exist at all should want that hardware to get free as fast as possible, and should be indifferent to whose badge is on it.

The honest weakness in that position. If the value moves to the OS layer, so does the competition — and it is Microsoft, Apple and Google, who ship *with* the hardware and own the defaults. That is a harder fight than competing on \$/GB/s, not an easier one. `os/`'s answer is structural: a model-as-OS with no legacy surface is the thing they cannot ship, because their franchise is the legacy surface. It is a real argument and an unproven one, and G6 is where it gets tested rather than asserted.

And what being unbuilt is worth. None of this costs anything but document edits, because nothing is sunk: no mask set, no board, no RTL, no inventory, no customer. That is not consolation, it is the asset — and the LPCAMM2 reversal on this page is the demonstration. An entire memory-system decision inverted, and the blast radius was one table, one entry in `decisions.md`, four rows in `system.py` and two in `bom.py`. The runtime, the format, the residency manager, the descriptor ring and the AI OS did not notice, because the memory module sits behind I5 and the rest was written not to care what it is. The same discipline is what lets the *whole chip* become optional without the plan collapsing: \$6 now reads with **Fork C as the base case**, and the silicon forks as options that need a market condition none of them currently has.

YFCE-M — the portable variant

"Mains-powered appliance" was a premise, not a requirement, and dropping it changes exactly one component: the memory modules. **LPCAMM2** is LPDDR5X on a socketed, compression-attached 128-bit module — 120 GB/s each at LPDDR5X-7500, up to 8533 at 1.05 V, with 9600 shown by Samsung and Lenovo at 96 GB and mass production expected 2027. Two modules is the shipping configuration (Crucial sells 32 and 64 GB modules at retail today); **four modules — 512 bits, 256 GB, 459 GB/s — is not shipping anywhere and is a signal-integrity question, not a purchase.**

configuration	bus	achieved	capacity now	2027	status
2× LPCAMM2-7500	256-bit	202 GB/s	128 GB	192 GB	shipping parts, proven topology
2× LPCAMM2-8533	256-bit	229	128 GB	192 GB	shipping parts
4× LPCAMM2-8533	512-bit	459	256 GB	384 GB	unproven at four modules
4× LPCAMM2-9600	512-bit	516	256 GB	384 GB	modules in MP ~2027

What it buys. Active memory power drops ~58% and standby ~80% against DDR5 SODIMM, and rather more against eight DIMMs. On the usual assumption of ~12 pJ/bit for a terminated DDR5 DIMM bus and ~5 for LPDDR5X, an 8B model at Q4 costs **432 mJ per token on DIMMs and 180 mJ on LPCAMM2** — 29 W against 12 W of DRAM traffic at 67 tok/s. A 99 Wh battery (the airline limit) is then about **4 hours generating at 25 W and 12 hours listening at 8 W**, and a voice-first machine spends almost all of its

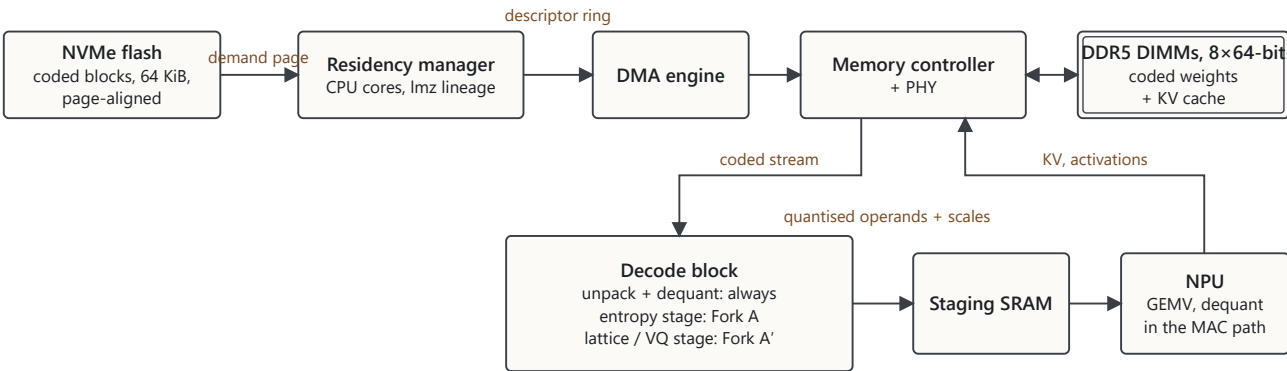
life listening, which is exactly where LPDDR's 80% standby saving lands. The board also shrinks: LPCAMM2 uses ~60% less area than SODIMM and needs no DIMM stub allowance, so the 14–16 layer motherboard becomes a portable's mainboard.

What it costs.

- **The node question gets worse, not better.** No 28 nm LPDDR5X-8533 PHY is known to exist — `system.py` already flags anything above LPDDR4X-4266 as not-mature-node. A portable probably buys a 16/12 nm program: the die is still trivial (~\$15–20 against \$5), but the mask set roughly triples and NRE goes from ~\$40M toward ~\$48M, moving the crossover from ~87k units to ~103k at pre-spike DRAM. This is E3/G2 (\$6) with a heavier weight on it.
- **The \$/GB argument that justified DIMMs does not survive.** It was 3× — and at August 2026 retail it has *inverted*: LPCAMM2 at \$8.26/GB (Crucial 64 GB, \$7.06 at launch) against DDR5 UDIMM at \$12–18.50/GB. On the pre-spike basis the DIMM still wins by 2.8×; on today's it loses. Both numbers are in `model/system.py`, on one basis, with the date attached, and `decisions.md` now carries the reversal.
- **Prefill on a battery.** §3's answer to prefill is either a bigger NPU or a MoE kernel; on a portable the bigger NPU also costs watts, which makes the MoE kernel the answer rather than an option.

What it does not change. The runtime, the format, the residency manager, the AI OS, the decode-block question (still Fork B unless G1 passes), the descriptor ring, and the invariant. YFCE-M is a memory-module choice and a power budget, not a different machine — which is the point of having written the design down as three layers with contracts between them.

2. System decomposition



Modules, what each one produces, and where it stands:

#	module	artefact	state
M1	Representation & format — the bytes on flash = in DRAM = into the decoder	format spec v0, reference encoder/decoder (golden model)	E1 done at 4-bit; format unspecified beyond Imz's block structure
M2	Encoder toolchain — BF16 checkpoint → YFCE archive	<code>yfce encode</code> , from Imz	Imz exists; the lossy half does not
M3	Residency manager — page-granular demand paging, expert/adapter swap, descriptor issue	part of the runtime	Imz mount/MappedArchive exist; no ring, no driver

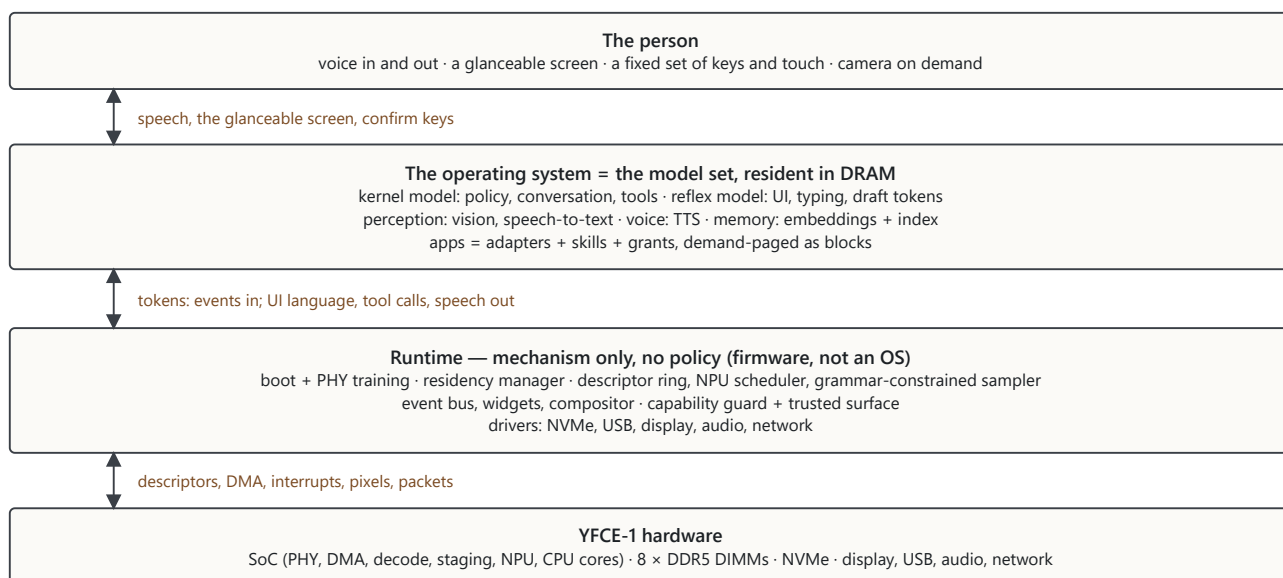
#	module	artefact	state
M4	DMA / descriptor engine — ring → DRAM reads → decoder slots	RTL	none
M5	Decode block — unpack/dequant always; entropy and lattice stages conditional	RTL + PPA	estimate only (<code>model/decoder.py</code>)
M6	Staging SRAM + NPU interface — operand tiles to the MAC array	RTL, sizing	16 MB assumed in <code>bom.py</code> , unsized
M7	NPU — GEMV at bus rate, prefill and encoders	licensed or designed	not started; TOPS is not its figure of merit for decode, and is for prefill
M8	Memory controller + PHY — 8–12 × 64-bit DDR5	licensed IP	E3: licensability at 28 nm unknown
M9	CPU cores, fabric, IO — run the runtime; NVMe, network; YFCE-P: display, USB 3, audio, camera	licensed IP	<code>bom.py</code> "misc"
M10	Board — 512/768-bit DIMM routing, power, thermal; YFCE-P: human I/O connectors	schematics, layout	none
M11	Runtime — bare-metal mechanism: boot, drivers, residency, NPU scheduling, sampler, event bus, widgets, capability guard	firmware image	none; \$3
M12	Model set — kernel, reflex, perception, voice, memory models; adapters as apps	archives	off-the-shelf models plus adapters trained in S9
M13	Interaction and tool contracts — speech, the UI language, the fixed touch/key set, the tool ABI	<code>os/spec/</code> : charter and first specs; grammar + renderer to come	\$3, \$4, S9
M14	Test & bring-up — BIST, host IF, boards	plans, BOMs	\$8

M1–M6 and M11–M14 are the project's own work. M7–M9 are bought. M10 is designed but its difficulty is routing, not novelty.

3. The AI PC — the model as the operating system

YFCE-P is YFCE-1 with human I/O and **no operating system**. Not an assistant running on Linux: there is no Linux, no Windows, no kernel, no processes, users, files, shell, desktop or applications anywhere the user can reach. The thing that owns *policy* — what is resident, what runs, what the screen shows, what happens on input — is a model. This is the argument of `architecture.md §3` ("a residency manager, not a POSIX layer") carried one layer up: the OS is not a purchased general-purpose kernel with the model as a tenant. The model is the OS; everything under it is mechanism.

It is **voice first**: the person speaks and the machine speaks back, with a glanceable screen that shows what it heard, what it is doing and what it needs. Touch and keyboard exist as a small fixed set the runtime owns — wake, confirm/cancel, a numeric pad for secrets, volume, tap/scroll on what is shown, transcript correction, an optional keyboard — and nothing else: no desktop, windows, menus or settings screens. The model set controls the apps and the system settings through the tool ABI. `os/README.md` is the charter; `os/spec/` holds the contracts.



What stays code, and why. A model decodes at tens of tokens per second. DDR5 training, DMA, interrupts, USB, pixel pushing happen at nanoseconds to milliseconds and must never be wrong. So under the model there is a **runtime**: firmware, not an OS. Single address space, static memory plan, one event loop; no processes, no users, no dynamic code — only weights and adapters are ever loaded, and those are data. It has no policy of its own: it does what the model's tokens say, within capabilities the user granted, and it owns the widgets' reactive behaviour (caret, scroll, pointer, audio playback) so the model is never in the keystroke path. Vendor firmware for PHY training and the boot ROM sit under it, as they would under any OS. Target size: ~50k lines of own code plus borrowed libraries (USB host stack, TCP/IP, TLS, image and audio codecs) — 100–150k lines in all, small enough for one team to own and audit. If the device set outgrows that, a verified microkernel (seL4-class) hosting the same drivers is the fallback: still nothing the user meets, but it is a kernel, and the baseline here is firmware.

The model set is the OS image. Resident together in 128 GB:

role	example	resident	job
kernel	70B-class dense at Q4, or a 120B-A5B MoE (\$1)	39–68 GB	policy, conversation, tool use, adapters
reflex	8B-class at Q4	4.5 GB	UI updates, typing assistance, draft tokens for speculative decoding of the kernel
perception	vision encoder (~0.4B), speech-to-text (~0.8B)	~2 GB	camera, imported images, microphone
voice	TTS (~0.5B)	~1 GB	speech out
memory	embedding model (bge-m3-class) + index	~1–2 GB	recall over the log and the user's things
KV cache	kernel 32k + reflex 32k	11 + 4 GB (dense kernel)	the process state; never re-prefilled
		≈ 60–90 GB	the rest is adapters and working set

Upgrading the OS is replacing archives; lmz's delta coding across checkpoints (a fine-tune differs little from its base) makes an update a small download.

Why the kernel wants to be a MoE. A dense 70B kernel gives 7.6 tok/s at 8 channels and prefills 4k tokens of a document in 29 s at 20 TOPS — an OS that answers in a minute. A 120B-A5B MoE at 4.5 bpw is 67 GB resident, reads 2.8 GB per token, gives **107 tok/s** on the same bus and prefills 4k tokens in ~2 s at 20 TOPS. Sockets buy the capacity that makes a MoE resident; bandwidth is spent only on active experts; expert swap at page granularity is what the residency manager is for. The DIMM thesis and the AI-OS point the same way. (Dense-kernel numbers with speculative decoding from the reflex model land around 15–20 tok/s and remain prefill-bound on documents.)

Speech is the primary output; the screen mirrors it. The model writes one short stream — a sentence or two, 15–30 tokens by policy — which the runtime speaks (phrase-streamed TTS) and shows. The reflex model acknowledges while the kernel works; barge-in stops speech within 100 ms; a spoken "undo" is a first-class command. Speech in is streaming ASR in the perception model, behind a wake word or push-to-talk that the runtime owns, with the microphone's state on an LED the model cannot drive.

The screen is a token stream. The model does not draw pixels. It emits a small declarative UI language — a versioned tree of text, lists, tables, images, fields, buttons and canvas primitives, with incremental updates — that the runtime renders at 60 Hz. **Tokens per screen update × tokens per second is the UI latency**, and it is a first-order design metric: at 67 tok/s a 40-token update takes 0.6 s and a 300-token screen 4.5 s; at 107 tok/s the 40-token update is 0.37 s. So the language is terse and diff-based — under voice it is further held to three glanceable regions (heard / doing / need) — the reflex model draws while the kernel thinks, and everything reactive is a widget. Input goes the other way: the fixed touch/key set, transcribed speech and encoded frames become tagged tokens in context. A grammar in the sampler (constrained decoding) makes a malformed UI tree or tool call impossible, not merely unlikely.

Apps are adapters. There are no applications. An "app" is a bundle of blocks — an adapter (LoRA or expert set), a skill document, tool grants, UI templates — installed into the same archive format and demand-paged by the residency manager when the kernel invokes it. Switching apps is a descriptor change, not a load. This is what expert/adapter swap in §2 exists for. **Settings are tools too** (`os/spec/tools.md`): a typed store in the runtime that the model reads and writes in tiers — everyday settings the reflex model changes on request (volume, brightness, do-not-disturb, timers, playback); consequential ones the kernel changes only after a spoken confirmation *and* the physical confirm key (network, pairing, power, updates, import/export, accounts, sharing); and a short list no model touches — microphone and camera mute (a hardware switch), factory reset, boot medium, and the capability grants themselves.

State and memory. No user-visible files. The machine's state is the persistent KV cache (context is never re-prefilled — that is the process state), an append-only log of interactions (what it remembers), and the user's things — documents, images, messages — as blocks in the archive, indexed by the memory model. Persistence is the block store; "save" is not a verb the user has.

Trust. The runtime is the trusted computing base; the model is not. Every input the model reads — a page, a message, an image, and under voice **any sound the microphone hears** — is a potential instruction, so prompt injection is this machine's malware and audio is its easiest delivery. Hence: tools are capabilities the user grants through a fixed **trusted surface** — a runtime-drawn prompt the model cannot draw over, plus **physical confirm keys** a sound cannot press — for every consequential action; speaker verification with liveness for the owner's voice; the machine's own playback subtracted from what it hears (echo reference); network egress and device access gated; every tool call logged; archives signed and the boot chain verifying the runtime and the model set. The model may be wrong; the runtime must not be.

`os/spec/trust.md` has the whole list.

Boot and power. Boot ROM → signed runtime (< 1 s) → reflex model resident (4.5 GB, under a second from NVMe) → usable; the kernel and the rest stream in behind it (~48 GB, ~8 s at ~6 GB/s). Suspend is DRAM self-refresh at single-digit watts; resume is instant with the KV cache intact.

Latency budget (8 channels, the reference configuration):

interaction	path	time
touch, scroll, volume, playback	widget in the runtime	≤ 16 ms, model not involved
wake → listening cue · barge-in → speech stops	runtime	≤ 200 ms · ≤ 100 ms
end of utterance → transcript on screen	streaming ASR (perception model)	≤ 300 ms
→ first spoken word of an acknowledgement	reflex model + streaming TTS	≤ 0.5 s
→ first spoken word of the answer	kernel + TTS: MoE · dense with speculation	≤ 1.5 s · ≤ 3 s
reflex response, first token	8B, incremental prefill of the new tokens only	~0.1 s
reflex 40-token UI update	8B at 67 tok/s	0.6 s
kernel 40-token answer	120B-A5B MoE at 107 tok/s · dense 70B at 7.6 (15–20 speculative)	0.4 s · 5 s (2–3 s)
a 4k-token document into the kernel	MoE at 20 TOPS · dense at 20 TOPS · dense at 100 TOPS	2 s · 29 s · 6 s
an image	vision encoder at 20 TOPS + ~256–1k tokens of prefill	tens of ms + as above
cold boot to usable · to full residency	reflex first · everything	~1 s · ~8 s

What it asks of the silicon (C3, YFCE-P variant). Display output (DP or HDMI PHY + controller + a small 2D compositor), USB 3 host, multi-channel audio in (mic array, PDM/I2S) and out with an **always-on low-power path** for wake-word, voice activity and echo cancellation that runs with the NPU asleep, GPIO for the confirm keys, mute switch and LED, optional MIPI-CSI, secure boot, and a larger NPU: prefill of long inputs is the weak flank of `architecture.md`, and the AI PC makes it central. A dense 70B kernel needs ~100 TOPS to feel acceptable on documents; a MoE kernel does not, and S8 decides between growing the NPU (+50–70 mm² for ~100 TOPS at 28 nm — about \$4 more per die, ~30 W) and choosing the kernel. Nothing here needs the decode block: **YFCE-P is a Fork-B product** — the bandwidth machine with the right software — and it survives Fork C on server hardware (\$8, T2 bare-metal).

What a user gives up. Legacy software, and a keyboard-and-mouse desktop. Nothing built for Windows or Linux runs here; the web arrives as content the model renders, not as a browser engine; hands do a few fixed things and the voice does the rest. Positioning has to say so as plainly as it says "no prefill story".

4. Interfaces — the contracts

The claim in `architecture.md` is that the value comes from owning all three layers. Owning them means writing down the seams, so that the parts can be built and simulated separately and still fit.

I1 — Format. Coded 64 KiB blocks, 4 KiB-aligned payload start, indexed by destination offset (lmz's chunk table). Each block: header (format id, tensor id, offset in tensor, plane lengths, table references, CRC32); planes in a fixed order — coded index plane, coded scale/min planes, raw planes; per-block **S = 8**

interleaved rANS states, 12-bit probabilities, 16-bit renormalisation, 32-bit states, 4-byte flush per state; frequency tables stored **once per tensor** and referenced by id, not repeated per block. Open choices S2 must settle: shared interleaved byte stream (lmz today; hardware needs a per-cycle prefix-sum over the 8 renorm decisions) versus one substream per state (+16–32 bytes per block, trivially parallel); which planes exist for each fork's representation. **The stream must be decodable with no knowledge outside the block and its referenced table.** Archives carry a signed manifest (§3, trust).

I2 — Descriptor ring. One descriptor per block: {coded address, coded length, destination (staging slot / stream id), format id, table id, tag}; completion queue with tag + status + CRC result; doorbell; a block is the unit of DMA, of decode, and of demand paging — the same 64 KiB throughout.

I3 — Decoded operand stream. The decoder emits *quantised* operands (packed 4-bit indices, or lattice indices, plus sub-block scales/mins) — never BF16. Dequantisation happens in the MAC path, as every GPU kernel does today. Emitting BF16 would put 1.1 TB/s into SRAM for nothing.

I4 — Fabric. Multiple 512-bit AXI-class ports between controller, DMA, decoder and NPU; 301 GB/s needs ~6 ports at 800 MHz. Bandwidth, QoS and refresh behaviour belong to S5.

I5 — Board ↔ package. Ball map, DIMM topology (1 DPC), power domains, reference clocks. Not touched until M8 is a real IP with a real ball-out.

I6 — Runtime ↔ hardware. The runtime's driver contract for I2 and for the NPU scheduler; page-fault → residency manager path; the runtime consumes I3 tiles through the staging SRAM.

I7 — Event and token bus. Inputs (key, pointer, touch, transcribed speech, encoded frames, network payloads, timers) become tagged token spans in the kernel or reflex context; output tokens are demultiplexed by a fixed grammar into UI-language updates, tool calls, speech, and memory writes. The grammar lives in the sampler; the model cannot emit outside it.

I8 — Interaction: speech and the UI language. Speech in (streaming ASR behind wake/push-to-talk) and out (phrase-streamed TTS) as the primary channel; the versioned declarative tree of §3 with incremental updates for the glanceable screen; the runtime's fixed touch/key set and widgets define all reactive behaviour; token cost per update and the voice latency classes are tracked metrics of the spec, measured in S9. `os/spec/interaction.md`.

I9 — Tool ABI, the "syscalls". memory (recall/store), residency (load adapter/model), apps (invoke, hand off, compose), network (fetch/send), device (show, speak, listen, capture, USB import), **settings (get/set/watch, in tiers)**, timers and power, self (compact context). Each call names a capability; capabilities are granted through the trusted surface, logged, and revocable; the model has no tool that changes a capability. `os/spec/tools.md`.

5. Decisions this blueprint proposes

To be ratified (or refused) in `decisions.md`. Each has a reason and a place in the campaign where it is checked.

D1 — Symbol rate is set by coded bytes per symbol, not by bus width alone. `model/decoder.py` sizes the block for BF16: 2 bytes per symbol, ~231 Gsym/s at 301 GB/s. A 4-bit-index format at 4.30 bpw carries 0.54 coded bytes per symbol, so the same bus demands **~561 Gsym/s**; a 3-bit family ~734; a 2-bit family ~1,050. Under decoder.py's own gate and SRAM constants that is 700 / 918 / 1,309 lanes at 800

MHz — **8.4 / 11.1 / 15.8 mm²**, not 2.6. Still small next to a 512-bit PHY, but three to six times the number in the docs, and it grows exactly where the ratio win grows. This is the quantitative form of "ratio trades against decode cost". Checked in S3/S4.

D2 — Parallelism across blocks, not across lanes inside a block. `architecture.md` says lane count is a format parameter fixed before tapeout. It need not be. Keep lmz's $S = 8$ states per block as the format constant; let hardware run **B block-decoder slots concurrently** (≈ 88 slots for 561 Gsym/s at 800 MHz, ≈ 105 for the 12-channel family at 1 GHz). The descriptor ring already delivers independent blocks; a 64 KiB block at 4.30 bpw is 122k symbols, 15.3k cycles per slot, 19 μ s — fine for a stream that is prefetched in order. Interleaving 1,024 states inside a block instead would cost 4 KiB of flush per 64 KiB block — 6.25%, i.e. the entire E1 gain. Slot count becomes an implementation choice, the format stops carrying a hardware constant, and the entry in `decisions.md` open item 4 moves to "Reversed" if S3 and S5 confirm. Checked in S2 (overhead), S3 (RTL), S5 (SRAM and ring depth).

D3 — The decode block emits quantised operands (I3). Dequant in the MAC path; no BF16 expansion into SRAM. Checked in S3/S6.

D4 — Format vo descends from lmz's page-mapped archive (64 KiB blocks, 4 KiB alignment, chunk table), extended by tables-by-reference, plane ordering, a signed manifest, and a format id per block so that Fork A and Fork B blocks share one container. Checked in S2.

D5 — The decode-block bar. The block earns its place if a representation is decodable at bus rate in $\leq 4 \text{ mm}^2$ and $\leq 1 \text{ W}$ at **28 nm** (S4 estimate) and reaches $f \leq 0.85$ against the best cheap-decode incumbent at the same operating point (Q4_K_M at 4-bit; Q3_K / IQ3 at 3; IQ2 / Q2_K at 2) at iso-quality on perplexity **and** at least one downstream task. Below 0.85 a faster DRAM grade buys the same for less risk; $f \leq 0.75$ is decisive. Checked in S1.

D6 — Campaign order. S1 first, because it is cheapest and can still kill the block; S5 regardless of fork, because a bandwidth machine needs its memory system modelled either way; S3/S4 start on the common path (DMA, unpack/dequant, staging) and only grow the entropy stage if G1 passes; S9 runs alongside, because the AI PC depends on none of G1.

D7 — Mechanism and policy are split at the runtime boundary, and there is no OS. The runtime is firmware: no processes, users, files, dynamic code or policy. The model set owns every decision a user can observe. Applies to YFCE-1 as much as to YFCE-P — the appliance drops Linux too. Checked in S9 (contracts) and S10 (bare metal).

D8 — Apps are adapters. Installed as blocks in the archive; demand-paged; switched by descriptor. No other application model exists. Checked in S9.

D9 — UI is an emitted, grammar-constrained token language; its token cost is a design metric. Reactive behaviour belongs to widgets in the runtime, never to the model. Checked in S9 (tokens per update, latency classes).

D10 — The runtime is the TCB; the model is untrusted toward hardware. Capabilities, trusted surface, logging, signed archives, verified boot. Checked in S9 (injection test set) and S10 (boot chain).

D11 — The AI PC needs the bandwidth machine, not the decode block. YFCE-P is a Fork-B product; the NPU, not the decoder, is what it grows, and the kernel model's shape (MoE vs dense) is chosen against the NPU in S8/S9.

D14 — The hardware envelope is assumed to commoditise; the product is the software.

Bandwidth, capacity and the mature-node die saving are all perishable (§1), and a chip program lands in 2029–30 against a curve that gets there first. So residency and the OS are the deliverables, hosts are bought, and silicon needs a positive market condition to start rather than a negative one to stop. Consequence: Fork C is the base case (§6), S2/S7/S9 are the spine (§7), G4 and G6 become the load-bearing gates, and D5's decode-block bar is now academic unless something changes. Checked continuously — this is the one decision that a market event can reverse, in either direction.

D12 — Voice is the primary channel; touch and keyboard are a small fixed set the runtime owns. The model cannot add gestures or keys; the screen is glanceable and mirrors speech; reactive behaviour is never the model's. Checked in S9 (voice latency classes, tokens per update, the task set done by voice with hands busy).

D13 — Settings are tools in tiers; some controls are hardware only. Everyday / consequential / hardware-only as in `os/spec/tools.md`; the model cannot lower its own guard. Checked in S9 (injection set incl. audio).

6. Gates and forks

gate	question	pass	where
G1 representation	does any cheap-decode representation clear D5?	$f \leq 0.85$, iso-quality incl. downstream, decoder within budget	S1 (+S4)
G2 bandwidth ceiling	is a DDR5-5600 (or $\geq$ LPDDR5-6400) PHY licensable at 28 nm — and an LPDDR5X-8533 PHY at <i>any</i> node this project can afford, for YFCE-M?	yes, with area/price in hand	E3 — procurement, §8
G3 memory system	does 8ch DDR5 with the DMA pattern achieve ≥ 0.80 of peak?	Ramulator2 ≥ 0.80 incl. refresh and KV traffic	S5
G4 system	does one-representation-end-to-end beat convert-and-materialise on the same box?	cold start, swap, footprint by predicted margins; tok/s ratio tracks $1/f$	S7 (PC), T1/T2 (board)
G5 silicon at all	is there a market condition under which a chip beats buying hosts — i.e. $\geq$ NRE crossover units and a durable advantage still standing when parts arrive in 2029–30?	business, not engineering; currently no on the second clause (§1)	outside this document
G6 AI-OS viability	can a model-as-OS carry an ordinary day of PC tasks by voice at these speeds, safely?	S9 task set: $\geq 90\%$ of tasks completed through the model alone, $\geq 80\%$ hands-free; end of utterance $\rightarrow$ first spoken word $p50 \leq 0.5$ s ack / ≤ 1.5 s answer; ≤ 40 tokens per screen update; zero unauthorised tool calls under the injection set including audio	S9, S10

G1–G4 and G6 are unchanged by any of the above — they are questions about representation, PHYs, memory systems, residency and the OS, and none of them cares who builds the host. Only G5 changed, and it changed from "can we sell enough" to "is there any condition at all".

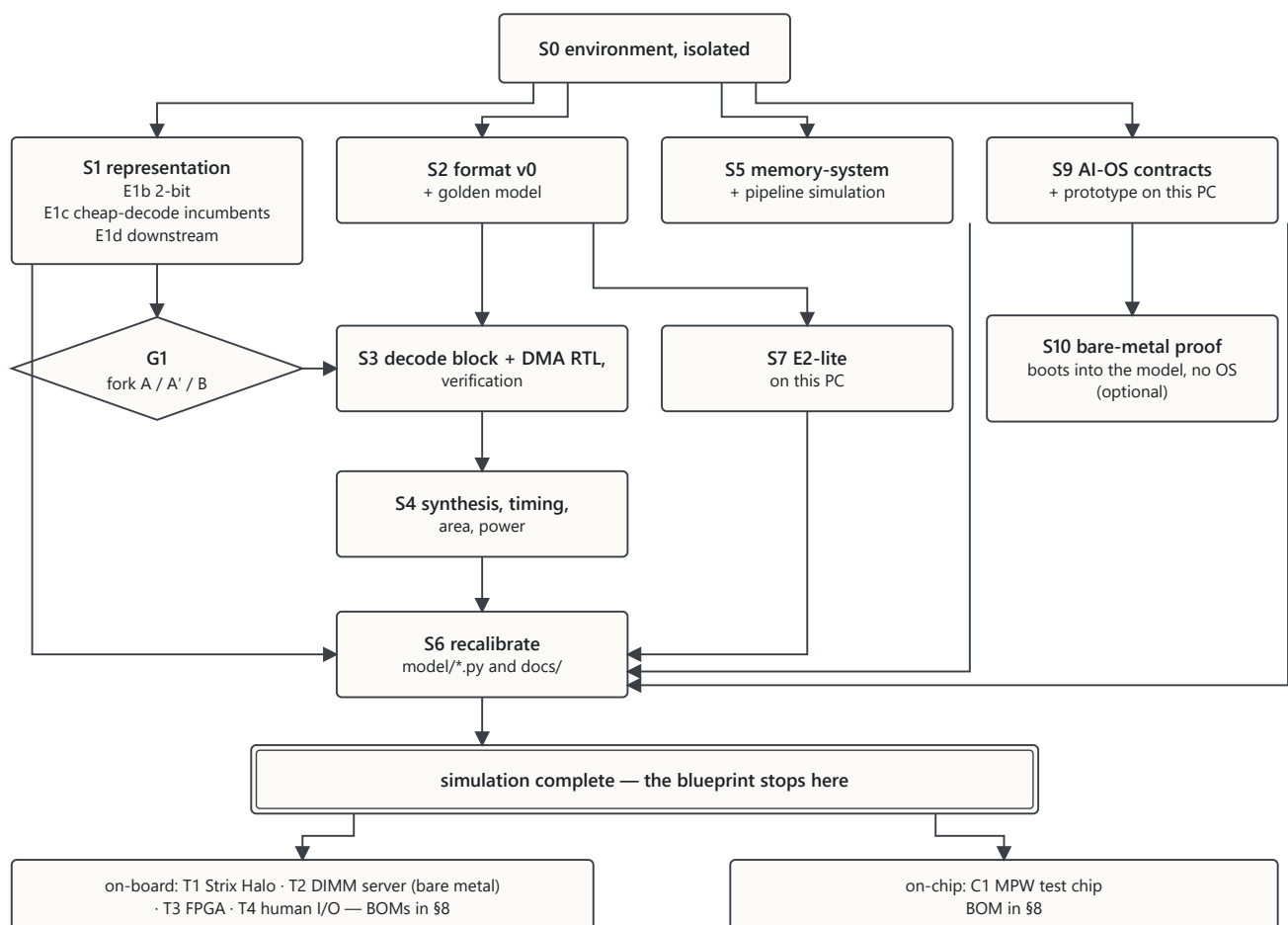
Forks, with the base case first:

- **Fork C — no chip: the base case.** The stack (M1–M3, M11–M13) on hosts somebody else builds, which is where the commodity curve in §1 points. The AI PC becomes a bare-metal runtime on a Spark-

class box, a Strix Halo box, an LPCAMM2 laptop or a DIMM server (§8, T1/T2), and gets faster and cheaper every year without a tapeout. Everything in `docs/simulation.md` except S3/S4 serves this fork directly; S2, S7 and S9 *are* this fork. It is selected by default and stays selected unless something below beats it.

- **Fork A — codec chip.** G1 passes with an entropy-coded family: decode block = unpack + entropy stage. Format carries coded planes.
- **Fork A' — lattice/VQ chip.** G1 passes with a lattice or codebook family (E8P-class, IQ-class, AQLM-class): decode block = unpack + table lookup + sign/scale apply; entropy stage optional on top. Rotation-based schemes (QuIP#, QuaRot) put their Hadamard on the *activation* side — once per token per layer, $O(d \log d)$ — which is the NPU's problem, not the decoder's; E1's inference that they are decode-expensive should be measured, not assumed (S1).
- **Fork B — bandwidth chip.** G1 fails. The SoC keeps PHY + controller + DMA + unpack/dequant + staging + NPU + CPU. Nothing distinctive in silicon; the product is a wide socketed-DDR5 board with the right software — and with §3, the right software is the whole product. Its competitor is a server CPU on the same DIMM topology (§8 T2) at 3–5× the BOM. G5 decides it. Forks A, A' and B are now **options against the base case, not destinations**: each needs G1 or G3 to produce something a commodity host cannot buy, *and* a G5 answer that survives §1's 2029–30 arithmetic. On today's evidence none of them has that, which is a finding, not a failure — it is what the gates were built to tell us, and it costs nothing because nothing was built.

7. Phase map



Under Fork C the campaign has a spine and a periphery. **S2 (format), S7 (residency, G4) and S9 (the AI OS, G6) are the product** and run regardless of every hardware question. S1 stays because bytes-per-token is the half of the roofline this project owns on anyone's hardware. S3/S4 (decode RTL and synthesis) become research against a chip that currently has no market condition — worth doing when there is slack, not worth blocking on. S5 stays because a bandwidth machine's memory behaviour has to be understood whoever builds it.

stage	answers	needs from PC	detail
S0	one isolated environment; nothing existing touched	disk ~60 GB	simulation.md \$S0
S1	G1 — representation and its decode cost	RTX 5080, ~5 min per (model, variant) point; downstream overnight	\$S1
S2	I1 as a spec and a bit-exact golden model	CPU	\$S2
S3	does the block decode at rate; how many slots; where it stalls	Verilator on 8 cores, unit runs in minutes, full-model overnight	\$S3
S4	area, fmax, power per slot; bracket 28 nm from open libraries	Yosys/OpenSTA, ≤ 16 GB RAM if scoped to a slot cluster	\$S4
S5	G3 — achieved bandwidth; SRAM and ring sizing; token latency	Ramulator2 + a discrete-event model, CPU minutes	\$S5
S6	the models and docs say what the simulations said	—	\$S6
S7	G4 at PC scale — the software architecture against stock llama.cpp	CPU + GPU, ext4 disk	\$S7
S8	prefill/NPU sizing and the kernel-model choice, on paper	—	\$S8
S9	G6 — the AI-OS contracts (I7–I9) as a running prototype: tokens per update, latency classes, injection set, an adapter trained as an "app"	RTX 5080 (8B + 3B + perception models fit 16 GB), QLoRA hours	\$S9
S10	the runtime boots this PC into the model with no OS on the machine (USB stick, CPU-only, reversible)	CPU AVX-512, ~10 tok/s on 8B; QEMU in the env for development	\$S10

Effort, one person: S0 1 wk · S1 3–4 · S2 2–3 · S3 6–8 · S4 2–3 · S5 3–4 · S6 1 · S7 3–4 · S9 5–6 · S10 4–8 (optional) → about seven months serial, four with S1/S5/S7/S9 in parallel to S2–S4. The campaign ends with a **simulation-complete package**: format spec, golden model, verified RTL with coverage, PPA brackets, calibrated models, a measured software baseline, and the AI-OS contracts with numbers behind them. That package is what the next-plans in §8 consume.

8. Beyond the PC — bills of materials and next plans

Not designed here. Under Fork C most of this section stops being a build plan and becomes a **shopping list of hosts and measurement rigs** — T1, T2 and T5 are things to buy and run the stack on, and only T3 and C1 are steps toward silicon. Prices are August 2026 street prices or quotes seen in public sources (listed at the end of this section), and DRAM in particular is in a price spike — verify before buying. Where "used" is noted the used market is the realistic route for a solo budget.

T1 — E2 on somebody else's 128 GB unified box

The competitor's memory architecture, purchasable. Proves the software system thesis (G4) at full scale; calibrates nothing about DIMMs. Two candidates now, and they answer different questions.

item	spec	est. price	note
Mini PC, Ryzen AI Max+ 395, 128 GB	GMKtec EVO-X2 / Framework Desktop / Beelink GTR9 Pro class	\$1,500–2,600	256-bit LPDDR5X-8000, ~215 GB/s measured — the cheap route, and the only measured number in \$1's table
RTX Spark system, 128 GB	any N1X OEM machine, autumn 2026	~\$2,899	the direct competitor: measures what a 256-bit LPDDR5X-9600 system <i>achieves</i> rather than quotes, which is the one number \$1 has to leave open, and gives a prefill baseline against 20 TOPS
NVMe, 2–4 TB, PCIe 4	Gen4 ×4	\$150–300	model archives + baselines
—		≈ \$1,700–2,900 (Halo) · ≈ \$3,050–3,200 (Spark)	

Next: install stock llama.cpp and the S7 stack; rerun S7's four measurements at 70B scale; publish the numbers against `model/system.py`'s prediction — and, on the Spark box, publish achieved GB/s and time-to-first-token beside this document's estimates for both. The AI-OS runtime cannot go bare-metal on either without a GPU driver, so it runs CPU-only or under a host OS there.

T2 — E2 on the machine's own memory topology (socketed, 8–12 channels)

A server CPU is the only thing on the market with a 512/768-bit DIMM bus. The CPU stands in for the NPU (AVX-512 GEMV reaches roughly 0.5–0.7 of peak); the DIMMs, PCB and controller are the real thing. This is the platform that calibrates S5's `BUS_EFFICIENCY` on actual 8-channel DIMMs — and, with the S10 runtime, **the first YFCE-P with no OS on the machine**: UEFI firmware boots the runtime, the runtime boots the model, AVX-512 does the GEMV at real DIMM bandwidth (8B ≈ 45–50 tok/s on 12 channels; 70B ≈ 5–6), the board's video output carries the UI, USB carries the input. Audio and camera wait for their drivers.

tier	build	est. price	what it tests
T2-lite (used)	EPYC 7003 "Milan" 8ch DDR4-3200: CPU \$300–500, board \$400, 8 × 32 GB RDIMM \$400–600, PSU/case/cooler \$250	≈ \$1,400–1,800	8-channel DIMM topology at 205 GB/s peak; the cheapest way to see a wide socketed bus behave; bare-metal runtime at ~30 tok/s on 8B
T2 (current gen)	EPYC 9004/9005 12ch DDR5-4800/6000: CPU \$600–1,100, board \$700–900, 12 × 16 GB RDIMM \$1,500–2,500 (2026 prices), PSU/case \$300	≈ \$3,100–4,800	the 12-channel family's bandwidth (~387 GB/s achievable) and DDR5 RDIMM behaviour; the bare-metal AI PC at 45–50 tok/s
T2-alt	Threadripper PRO 7955WX + WRX90, 8 × 16 GB DDR5-5200	≈ \$4,500	8ch DDR5 in a workstation form; costs more than EPYC for less width

Next: STREAM and a weight-streaming microbenchmark first (S5 calibration); then the S7 stack; then compare against T1 on identical models — this is the "sockets vs solder" decision measured rather than modelled. Then the S10 runtime image on a USB stick.

T3 — FPGA prototype of the decode path

RTL from S3 on real memory. Tiered because the question changes with the board: functional correctness needs almost nothing; decode-at-bandwidth needs HBM.

tier	board	memory	est. price	proves
T3-a functional	Artix-7 / Kintex-7 dev board (Nexys Video, KC705 used)	DDR3, ~12 GB/s	\$300–1,000	RTL runs on silicon; DMA ring and driver against a real bus; Vivado free edition
T3-b mid	Kintex UltraScale+ (KCU105/KCU116, Alinx AXKU040-class)	DDR4 64-bit, ~19 GB/s + PCIe host	\$1,000–3,000	slot array at scale, PCIe XDMA host link, real block traffic
T3-c bandwidth- class	Alveo U50 (HBM2 316 GB/s) or U55C (460 GB/s)	HBM	U50 \$1,000–3,000 used/new; U55C \$3,000–5,500	decode at bus rate: the S3 slot array fed at 300+ GB/s, measured, with power

Next: port S3 RTL (Verilator-clean SystemVerilog) to Vivado; wrap with the vendor memory controller and XDMA; drive from the S2 golden encoder over PCIe; measure symbols/s per slot, achieved memory bandwidth, and stalls against S5's prediction. T3-c is the first point at which "decodes at rate" becomes a measurement instead of an estimate.

T4 — Human I/O kit for the AI PC

What T2 (or this PC, S10) needs to become a machine a person sits at. Mostly things already on a desk.

item	est. price	note
Monitor (DP/HDMI, touch if possible), USB keyboard	\$150–300 if not owned	UEFI GOP and HID: usable from the first bare-metal build
USB microphone array (ReSpeaker-class) and speaker, USB webcam	\$60–150	voice-first needs the array first; USB audio-class and UVC drivers in the runtime are early milestones now
Confirm/cancel keys (a USB button box) and a mute switch in the mic path	\$15–40	the trusted surface, physically
USB 3 stick, 128 GB	\$15	the runtime image + model archives; the S10 boot medium
Wi-Fi/BT M.2 module (for a YFCE-P board)	\$15–30	not needed on T2 (wired)
—	≈ \$100–560	

T5 — LPCAMM2, measured (the portable variant's only untested part)

YFCE-M rests on two numbers nobody here has measured: what an LPCAMM2 system *achieves* against its peak, and what it costs in watts at idle and under streaming load. Both are answerable on a laptop that already ships with the modules — no board design, no silicon.

item	est. price	what it answers
A 2× LPCAMM2 laptop (Lenovo ThinkPad P1-class, or an Intel Panther Lake platform; borrowed or used)	\$1,200–2,500, or nothing if borrowed	achieved fraction of 256-bit LPDDR5X peak under STREAM and under weight streaming — the BUS_EFFICIENCY that S5 assumes at 0.84
A spare 64 GB LPCAMM2 module	\$450–665	capacity scaling, and whether \$/GB is really below DDR5 at the till

item	est. price	what it answers
Inline DC power meter, or the platform's own energy counters	\$30–100	idle, listening and generating watts → the 432 vs 180 mJ/token estimate, and the battery claim
—	≈ \$500–3,300	

Next: STREAM and a weight-streaming microbenchmark, then the S7 stack, then the same runs with the machine on battery. If achieved efficiency comes in near 0.84 and the energy numbers hold, YFCE-M's case is arithmetic; if either misses, the portable variant is where it shows first and cheapest. The four-module question — 512 bits of LPCAMM2 — is *not* answered here and cannot be bought: it needs T3-class board work or a partner platform.

C1 — MPW test chip of the decode block

Only if G1 passes. A few mm² on a 28 nm multi-project shuttle: the slot array (as many as fit), the DMA engine fed from on-chip pattern SRAM and a BIST loop (no package can deliver 300 GB/s off-chip), a PLL, and a slow host interface (SPI/JTAG/UART). Measures fmax, area and power per slot on real silicon — the numbers S4 can only bracket.

item	est. price	note
Shuttle seat, TSMC 28 nm via Europractice / Muse / CyberShuttle, 3–5 mm ²	€12–14k per mm ² academic; commercial rates higher — budget €60–120k	typical 2–5 mm ² 28 nm project quoted at \$26–80k academic incl. dies
Foundry PDK, standard cells, memory compiler access	NDA; often bundled	required for real S4 numbers even before tape-in
Packaging, small batch (QFN/BGA)	€5–10k	
Test board, host FPGA (reuse T3-a/b), bench PSU/scope/JTAG	\$5–15k	
—	≈ \$100–200k, 6–9 months from tape-in to parts	

Cheaper existence proof: **IHP SG13G2 130 nm** on an open-PDK shuttle (Tiny Tapeout-class, €1–5k) proves the RTL-to-silicon flow with the same open tools S4 uses; it does not prove rate.

C2 — PHY: E3 procurement (start now; costs emails)

Not a chip of ours. Request from IP vendors and foundries: DDR5-4800/5600 and LPDDR5-6400 PHY + controller availability at TSMC 28HPC+, UMC 28, SMIC 28, GF 22FDX — and, for YFCE-M, **LPDDR5X-8533 PHY at 16/12 nm**, which is where that grade actually exists and what a portable therefore costs in NRE; area per 64-bit channel; hardened vs delivered; licence, royalty, and support terms; whether a vendor test chip exists. Vendors to ask: Synopsys, Cadence, Rambus, Alphawave, Innosilicon, Analog Bits, M31, Silicon Creations (PLLs), and the foundries' own IP catalogues. The answer sets G2 and the largest single NRE line in `model/bom.py`. For YFCE-P, add to the same round: DP/HDMI and USB 3 PHYs at the same node.

C3 — Full SoC and reference board

`model/bom.py`: ~\$40M NRE, ~107 mm², crossover ~87k units at \$2.50/GB DRAM — **~28k units at 2026's ~\$10/GB** (BOM rises to ~\$1,420; the chip's share of the machine shrinks while the machine gets dearer). The YFCE-P variant adds display, USB 3, audio, camera and secure-boot IP (a few \$M of NRE),

and +50–70 mm² if the NPU grows to ~100 TOPS (~\$4 per die at 28 nm, ~30 W). Reference board BOM lines beyond bom.py's aggregates: FCBGA SoC; 8 × DDR5 DIMM connectors (~\$2–3 each); VRMs for SoC and 8 DIMMs; PMIC; clock generator; NVMe M.2; 2.5GbE PHY + magnetics; USB; ~150 W PSU; chassis and thermal (~\$45 in bom.py); YFCE-P: DP/HDMI connector and level shifters, USB 3 hub, audio codec + jacks, Wi-Fi/BT M.2 (E-key), camera connector — roughly +\$25–45. Not planned further until G1–G6 are answered.

Sources consulted for the prices above (August 2026): [TweakTown — AMD Ryzen AI Halo dev kit \\$3,999](#) · [Liliputing — 128 GB Ryzen AI Max+ 395 mini PCs](#) · [Compute Market — Strix Halo mini PCs, July 2026 listings](#) · [modelfit — Strix Halo 128 GB price](#) · [RAM Price Index — DDR5 \\$/GB](#) · [datacenterdisk — 64 GB DDR5 RDIMM \\$/GB](#) · [TechRadar — DDR5 price forecast 2026](#) · [Silicon Analysts — MPW costs by node](#) · [Europractice — 2026 schedules and prices](#) · [PCBSync — Alveo U55C](#) · [PCBSync — Alveo U50](#) · [AMD — Alveo U55C](#)

9. Risks

risk	effect	mitigation / where it is watched
No cheap-decode representation clears D5	Fork B; nothing distinctive in silicon	S1 tests incumbents (IQ, imatrix, lattice) instead of only inventing; D5 fixed before results; the AI PC (D11) does not depend on it
No DDR5 PHY at 28 nm	ceiling 229 GB/s; compression becomes load-bearing at exactly the moment G1 says it is weak	C2 now; the family definition keeps a 22FDX / LPDDR5 branch open
DRAM price spike persists (2026: \$9–18/GB consumer, RDIMM doubled)	BOM ×3, retail price up, "sockets 3× cheaper than solder" needs re-measuring at current prices	S6 puts \$/GB in as a scenario, not a constant
Symbol rate under-estimated in current docs (D1)	decode block 3–6× larger than quoted	D1; S3/S4 measure per-slot cost
Format bakes in a hardware constant	format break on the next bus width	D2 block-parallel; S2 measures overhead
Open-library PPA misleads about 28 nm	wrong area/power going into the gate	S4 brackets with two libraries and reports the bracket, not a point; C1 or a foundry PDK closes it
WSL2 confounds S7 I/O measurements	cold-start numbers not transferable	S7 reports relative numbers on ext4 only; T1/T2 give absolutes
The model is not good enough to be an OS at 8–120B	G6 fails: tasks need a person to babysit, or tool calls misfire	S9 measures a task set, not a demo; reflex/kernel split; adapters trained for the UI and tool grammar; grammar-constrained sampling
Prompt injection as an OS-level attack	the model executes an attacker's instructions with the user's capabilities	D10: TCB in the runtime, trusted surface, gated egress, logging; the injection set is a G6 criterion, not a note
UI latency = tokens × tok/s	a slow, talkative UI language makes the machine feel broken	D9: terse diff-based language, widgets for everything reactive, reflex model draws; token cost tracked in S9

risk	effect	mitigation / where it is watched
Audio injection and voice cloning	a sound commands the machine with the owner's capabilities	D13 tiers; physical confirm keys for consequential actions; speaker verification with liveness; echo reference; the audio injection set is a G6 criterion
ASR errors become wrong actions	"call Ann" becomes "call Dan"	transcript shown before consequential actions; confidence-gated confirmation; spoken undo; measured in S9's task set
A listening machine	privacy, and the perception of it	microphone open only after wake, LED the model cannot drive, hardware mute, local by default, the log reviewable by voice
Prefill on long inputs (the weak flank, now central)	documents and images take tens of seconds on a dense kernel at 20 TOPS	MoE kernel (§3); NPU 60–100 TOPS as an option costed in S8
The bare-metal runtime grows into an OS	driver surface (USB, network, TLS, codecs) becomes unownable	staged device set; borrowed libraries; seL4-class fallback stated in §3; S10 measures what UEFI must provide
No legacy software	market: a PC that runs nothing existing	positioning as a new category, stated in §3; the network tool renders content, not apps
One-person bandwidth	S3 (RTL) and S9 (runtime) dominate the calendar	S1/S5/S7 run in parallel; S3 scoped to the common path first; S10 optional
The envelope commoditises — RTX Spark now at ~\$2,899, and \$1,000–1,500 class hardware by ~2028	every hardware advantage this project could hold is perishable, and a chip program lands after the curve does	D14: the product is residency and the OS, hosts are bought (§6 Fork C); the risk to manage is being <i>late to the software</i> , not being beaten on GB/s
The OS layer is contested by the platform owners	Microsoft, Apple and Google ship with the hardware and own the defaults; a better local OS can lose on distribution alone	stated in §1 rather than wished away; <code>os/</code> 's structural claim (their franchise is the legacy surface) is tested by G6, not asserted
The 4-module LPCAMM2 configuration is unproven	YFCE-M's 512-bit, 256 GB, 459 GB/s headline is the one number with no shipping precedent	§1 marks it as a signal-integrity question, not a purchase; 2 modules is the fallback and still ships; T5 measures 2, T3-class work is needed for 4
A portable needs a PHY grade that does not exist at 28 nm	YFCE-M buys a 16/12 nm program: NRE ~\$40M → ~\$48M, crossover ~87k → ~103k units	C2 asks for LPDDR5X-8533 PHY pricing at 16/12 nm in the same round as the 28 nm DDR5 request
Volume (G5)	decides silicon regardless of everything above, and now against an incumbent with distribution	stated, not solved

10. What this blueprint deliberately leaves out

- Any RTL, spec text, runtime code, or model training — those are S2/S3/S9 deliverables, and the simulation plan says how they are built and checked.
- The board layout, the package, and the SoC floorplan beyond bom.py's aggregates. They wait on M8 being a real IP.
- Prefill beyond a sizing table (S8). Compute-bound and unaddressed, as `architecture.md` says — now with the kernel-model choice hanging on it.

- The UI language and tool ABI as text. §3 and §4 fix what they are and what they must cost; §9 writes them and measures them.
- Multi-user, real-time guarantees beyond widget reactivity, and running anyone else's software.
- A go/no-go on tapeout. That is G1–G6, and this document only says how each gets answered and in what order.

Appendix — numbers used above

Achieved bandwidth at 0.84: 8ch DDR5-5600 **301 GB/s**, 12ch **452**, 8ch DDR5-4800 **258**, 512b LPDDR4X-4266 **229**; Strix Halo 215 measured; RTX 5080 806 (960 peak).

Symbol rate = achieved bandwidth ÷ coded bytes per symbol, at 301 GB/s: BF16 exponent-coded (1.31 B/sym) 231 Gsym/s · 4-bit + entropy (0.54 B/sym) 561 · 3-bit family (0.41) 734 · 2-bit family (0.29) ~1,050. At 452 GB/s multiply by 1.5.

Slots at S = 8: 561 Gsym/s → 88 at 800 MHz; 841 Gsym/s → 105 at 1 GHz. Block: 64 KiB at 4.30 bpw = 122k symbols = 15.3k cycles = 19 μs at 800 MHz. Flush overhead per 64 KiB block at 4 B/state: S=8 0.05% · S=256 1.6% · S=1024 6.25%.

BOM by DRAM price (`model/bom.py` , 8ch DDR5-5600, 128 GB): \$2.5/GB → \$462, crossover 86.7k units · \$5 → \$782, 51k · \$10 → \$1,422, 28k · \$15 → \$2,062, 19k. Die: 107 mm² \$5.4 · 170 mm² \$9.3 · 190 mm² \$10.6 (28 nm, `bom.py` yield).

AI PC: KV per token, GQA 8 heads × 128, 16-bit — 70B (80 layers) 320 KiB, 32k = 10.7 GB, 128k = 43 GB; 8B (32 layers) 128 KiB, 32k = 4.3 GB. MoE 120B-A5B at 4.5 bpw: 67.5 GB resident, 2.81 GB active → 107 / 76 / 161 tok/s at 301 / 215 / 452 GB/s; prefill 2,000 tok/s at 20 TOPS. Dense 70B: 143 tok/s prefill at 20 TOPS, 714 at 100. UI: 40 tokens = 0.6 s at 67 tok/s, 0.37 s at 107; 300 tokens = 4.5 s at 67. Boot: 4.5 GB reflex ≈ 0.75 s and 48 GB set ≈ 8 s at 6 GB/s.

Field, August 2026 (§1): RTX Spark N1X — N3, 256-bit LPDDR5X-9600, 128 GB soldered, ~300 GB/s quoted (~252–258 at this document's 0.84 derating), ~1 PFLOP, ~\$2,899, autumn 2026; DGX Spark \$4,699. LPCAMM2: 128-bit socketed LPDDR5X module, 120 GB/s at 7500, up to 8533 at 1.05 V, 9600 sampling at 96 GB (MP ~2027); 64 GB module \$451.99 launch MSRP, \$528–665 August 2026 retail (\$8.26–10.39/GB) against DDR5 UDIMM \$12–18.50/GB. Memory energy at ~12 pJ/bit (DDR5 DIMM) and ~5 (LPDDR5X): 8B Q4 costs 432 vs 180 mJ/token, 29 W vs 12 W at 67 tok/s; a 99 Wh battery gives ~4 h generating at 25 W and ~12 h listening at 8 W. Prefill of 4k tokens: 28.7 s (70B dense, 20 TOPS), 2.05 s (120B-A5B, 20 TOPS), 0.57 s (70B dense, 1 PFLOP).

PC envelope this campaign is sized to: Ryzen 7 9800X3D (8C/16T), 24 GB RAM in WSL2 (31 GB host), RTX 5080 16 GB, CUDA 13.2 toolkit present, ~700 GB free on the WSL ext4 disk, models on `/mnt/d/Models`. Details and constraints in `docs/simulation.md`.

Simulation plan

The PC-side campaign from `blueprint.md` §7, stage by stage: what each stage asks, how it is built, what it costs on this machine, and how its result is read. It ends at "simulation complete"; the board and silicon steps that follow are BOMs and next-plans in `blueprint.md` §8. S9 and S10 are the AI PC's stages (blueprint §3): the model-as-OS contracts as a running prototype, and — optionally — this PC booting into the model with no OS.

The envelope

Everything here is sized to the machine this repository lives on:

resource	available	consequence
CPU	Ryzen 7 9800X3D, 8C/16T, 96 MB L3	Verilator, Ramulator2, Yosys run comfortably; RTL sims parallelise across seeds, not threads
RAM	24 GB to WSL2 (31 GB host, <code>.wslconfig</code>), 8 GB swap	an 8B BF16 checkpoint (16 GB) fits once on the CPU; never two copies; P&R scoped to a slot cluster
GPU	RTX 5080, 16 GB, sm_120	8B-class models only; embeddings stay on CPU or use <code>accelerate</code> offload; 70B is out of reach here
CUDA	<code>/usr/local/cuda-13.2</code> (nvcc 13.2, driver 595)	targets sm_120; the apt <code>nvidia-cuda-toolkit</code> (12.4) cannot and must not be installed
Disk	~700 GB free on the WSL ext4 disk; models on <code>/mnt/d/Models</code> (NTFS via 9P, slow)	copy working models to ext4; do all I/O measurements on ext4
OS	Ubuntu 26.04 in WSL2 (WSLg gives a window), no Docker	storage timings are Hyper-V-virtualised — S7 reports relative numbers; S9's renderer draws into a WSLg window or a local browser tab
Firmware	UEFI PC; boots USB media	S10 can boot the runtime from a stick without touching either installed OS; QEMU + OVMF inside the env for development
Existing envs	<code>torch_env</code> , <code>exatrkx</code> (torch 2.12 cu130, transformers 5.8), system Python 3.14	read-only for this campaign — nothing is installed into or upgraded in them

Isolation rule. One new conda environment (`yfce`), one source-build prefix (`~/opt/yfce`), one tools checkout directory (`~/src/yfce-tools`). No `apt install` is required; where an apt package would do, the conda-forge equivalent inside `yfce` is listed first and apt is the alternative for the user to choose. Nothing in `/usr`, `/usr/local/cuda*`, or the existing envs is modified.

S0 — Environment

component	how	size	used by
Python 3.12 env <code>yfce</code>	<code>conda create -n yfce python=3.12</code>	0.5 GB	all
torch (cu13x wheel), transformers, safetensors, accelerate, numpy, scipy	pip inside <code>yfce</code> ; <code>--index-url</code> for the cu130 wheel matching the driver	6 GB	S1, S7

component	how	size	used by
lm-eval (<code>lm_eval</code>), <code>gguf</code> (llama.cpp's Python package), hypothesis, pytest, simply, cocotb	pip inside <code>yfce</code>	small	S1, S2, S3, S5
llama.cpp	<code>git clone</code> into <code>~/src/yfce-tools</code> , <code>cmake -B build -DGGML_CUDA=ON -DCMAKE_CUDA_COMPILER=/usr/local/cuda-13.2/bin/nvcc</code> , build in-tree, no install	2 GB	S1 (real llama-quantize, llama-imatrix, llama-perplexity), S7 (baseline)
lmz	submodule <code>lmz/</code> (<code>git submodule update --init</code>), <code>lmz/lmz-cli</code> ; the <code>gpu-decoder</code> branch on its origin is the natural seed for S7-b	—	S2, S7
Verilator, Icarus, Yosys, GTKWave	<code>conda install -c conda-forge verilator iverilog yosys gtkwave</code> (alt: apt has verilator 5.032, iverilog 12, yosys 0.52)	1 GB	S3, S4
OpenSTA	source build, <code>-DCMAKE_INSTALL_PREFIX=~/opt/yfce</code> ; <code>tcl/swig/eigen</code> from conda-forge	0.5 GB	S4
Open libraries: Nangate45 (from OpenROAD-flow-scripts <code>platforms/</code>), ASAP7, sky130 HD, IHP sg13g2	clone into <code>~/src/yfce-tools</code>	3 GB	S4
OpenROAD (optional, tier 3 of S4)	prebuilt Ubuntu 24.04 release unpacked under <code>~/opt/yfce</code> (<code>dpkg -x</code>), or the litex-hub conda channel — verify either works on 26.04; skip if not	2 GB	S4
CACTI 7	source build in <code>~/opt/yfce</code>	small	S4
Ramulator 2.0 (and/or DRAMsim3)	source build (C++20) in <code>~/opt/yfce</code>	small	S5
Models	Llama-3.1-8B-Instruct BF16 (present), Q8_0 GGUF (present), Ministral-8B (present); generate Q4_K_M / Q3_K_M / Q2_K / IQ4_XS / IQ3_XXS / IQ2_XS GGUFs; optional second family (Qwen3-8B)	40–60 GB on ext4	S1, S7
S9 model set	a 3B-class reflex model (Q4), whisper-small/turbo, a ~0.4B vision encoder, a small TTS, bge-m3 (present) — GGUF or safetensors	~6 GB	S9
peft + a 4-bit training backend (bitsandbytes or torchao — whichever ships a cu13x wheel at install time), datasets	pip inside <code>yfce</code>	small	S9 (QLoRA of the UI/tool-grammar adapter)
qemu-system-x86_64 (conda-forge) + an OVMF firmware image; gnu-efi source vendored into <code>~/src/yfce-tools</code> and built in-tree	inside the env / tools dir; nothing system-wide	0.5 GB	S10

Half a day plus build time (llama.cpp CUDA build ~15 min on 8 cores). Verify at the end: `python -c "import torch; torch.cuda.get_device_capability()" → (12, 0); verilator --version; Yosys -V; llama-quantize --help`.

S1 — Representation: G1

Question. Does any representation that a mature-node fixed-function block can decode at bus rate reach $f \leq 0.85$ against the best cheap-decode incumbent at the same operating point, at iso-quality on perplexity and one downstream task (blueprint D5)?

E1 tested one family (non-uniform scalar + entropy) and it lost; its 3-bit point (6.22%) is already in `experiments/e1_representation/RESULTS.md`. This stage closes the 2-bit point cheaply and then tests the family that actually beats Q4_K_M — but scores it on decode cost as well as ratio.

E1b — 2-bit, the E1 family. `run.py --levels 4` as it stands. Expected ~8%, far below the bar; ~1 hour; closes `decisions.md` open item 1.

E1c — cheap-decode incumbents. Quantise Llama-3.1-8B with llama.cpp's own quantisers: Q4_K_M, Q3_K_M, Q2_K, IQ4_XS, IQ3_XXS, IQ3_S, IQ2_XS, IQ2_XXS, with and without an importance matrix (llama-imatrix on a wikitext-train slice). The IQ family is a lattice codebook (E8-class, 256–1024-entry 8- or 4-dim grids plus sign patterns) — decode is a table lookup, a sign apply and a scale: cheap in hardware, and the best ratio the incumbents have. Evaluate every variant **inside E1's harness**, not with `llama-perplexity`: dequantise each GGUF tensor with `gguf-py`, load into the transformers model in place of the projection weights, run `ppl.py` on the same 250×512 -token windows. That keeps the numbers on the same axes as E1's table. `llama-perplexity` on the same files is a cross-check only. Bits per weight are computed from the file, per tensor class, not from the type's nominal figure.

E1c-r — rotation + uniform grid. One QuaRot-class candidate in the harness: block-diagonal random Hadamard on input/output dimensions, GPTQ ($128 \times 2k$ calibration tokens, ~1–2 h on the 5080 layer-by-layer), uniform 3- and 4-bit. Its decoder is the plain uniform dequant; the transform lands on the activation side, once per token per layer, $O(d \log d)$. This is the measurement `RESULTS.md` inferred rather than made.

Entropy on top. E1's entropy accounting (`quantize.py` order-0 and conditioned) applied to the IQ and E1c-r index and scale planes, to see whether any lossless margin survives on top of a good lossy stage.

Decode-cost score, per variant. Coded bytes per symbol $\rightarrow$ Gsym/s at 301 and 452 GB/s (blueprint D1); table bits; operations per weight (lookup, sign, multiply-add, transform); slots at $S = 8$; projected area and power from `model/decoder.py` (constants replaced by $S4$ as it lands). Reported as a column next to bpw and perplexity — a candidate is scored on $(\text{quality}) \times (1/f)$ **subject to** the D5 budget, as `experiments/README.md` already demands.

E1d — downstream. lm-eval on the top three candidates and their baselines: hellaswag, arc_challenge, winogrande, mmlu (5-shot). Fake-quant model through lm-eval's HF backend with `accelerate` offloading the embeddings. 8–16 GPU-hours, overnight. Then the top candidate on Ministral-8B, so the result is not a Llama artefact.

Budget. ~5 min per (model, variant) perplexity point (77 s bf16, 200–270 s quantised, measured in `e1.log`); ~30 points $\approx$ 3 h; GPTQ 1–2 h; downstream overnight; total $\approx$ 2 GPU-days plus 3–4 weeks of work.

Outputs. `experiments/e1c_incumbents/{CONDITIONS.md, RESULTS.md, *.json}` in E1's format; the perplexity-vs-bpw table with a decode-cost column; the G1 verdict against D5, and which fork it selects.

Read. Pass → Fork A or A', and S3 grows the corresponding stage. Fail → Fork B, S3 stays on the common path. Either way S5 and S7 proceed.

S2 — Format v0 and the golden model

Question. What exactly is on the wire (blueprint I1), and can it be encoded and decoded bit-exactly by a model the RTL is checked against?

Deliverables.

- `format/SPEC.md` — vo: container (lmz page-mapped lineage: 64 KiB coded blocks, 4 KiB-aligned payloads, chunk table keyed by destination offset), block header (format id, tensor id, offset in tensor, plane lengths, table ids, CRC32), plane order, table-by-reference section, rANS parameters ($S = 8$, 12-bit probabilities, 16-bit renormalisation, 32-bit states, 4-byte flush), the descriptor record (I2), and one format id per fork (raw Q4_K passthrough is a valid format so Fork B uses the same container).
- `format/yfce/` — Python package: encoder, decoder, block index, descriptor generator, CRC; both stream layouts (shared interleaved as lmz today; one substream per state) behind a flag; C fast path by reusing `lmzcore.c`'s rANS for the lmz-compatible layout.
- `format/vectors/` — golden vectors: real Llama tensors of each shape class (attention, MLP, small); synthetic edge cases: single-symbol alphabet, frequency-1 symbols, empty and raw-fallback planes, maximum block, table change between consecutive blocks; each with expected decoded output and descriptors.

Decisions settled by measurement here. Per-state substreams vs shared stream (overhead is $\leq 0.05\%$ either way; choose per-state unless S3 shows the 8-way prefix-sum is free); tables by reference vs per block (per-block tables cost $\sim 0.8\%$ of a 64 KiB block — most of the E1 gain — so by-reference unless S3 shows table reload stalls dominate); 64 vs 128 KiB blocks (random access granularity vs header overhead); which planes each fork's format id carries.

Tests. Round-trip on all vectors; cross-decode against lmz for the compatible layout; hypothesis property tests on random planes; a full 8B archive encoded and decoded (Python rANS ~ 10 MB/s → ~ 10 min with the C path, hours without — either is fine).

Budget. CPU only; 2–3 weeks.

S3 — Decode block and DMA engine: RTL and verification

Question. Does a block-parallel decoder (blueprint D2) sustain one symbol per state per cycle under realistic DMA behaviour, how many slots does the bus need, and where does it stall?

Language and tools. SystemVerilog, synthesizable subset that is Verilator-clean, Yosys `read_verilog -sv`-clean and Vivado-clean (T3 will want it unchanged). Verilator as the primary simulator, Icarus for a periodic second opinion on the unit level, cocotb for testbenches, GTKWave for waves.

Microarchitecture (parametric).

- `slot` — $S = 8$ rANS state machines: 32-bit state; 12-bit slot extract; symbol lookup — table organisation is a parameter with three options to be costed in S4: `tree16` (16-entry cumulative-frequency compare tree, no SRAM: right for 4-bit index alphabets), `two-level-256` (16 groups $\times$ 16), `direct-4096` (4096-entry symbol map, 4 KiB per copy — simplest, largest); update `x = freq.(x >> 12) + slot - cum`; 16-bit

renormalisation from the slot's input FIFO, per-state or via an 8-wide prefix-sum on the shared stream (parameter). Then the **unpack/dequant stage** (present in every fork): symbols $\rightarrow$ packed 4-bit indices (or lattice indices) + sub-block scales/mins $\rightarrow$ I3 tiles.

- `slot_array[B]` — dispatcher assigns descriptors to free slots; per-slot input FIFO fed by the DMA (bursts 64 B–4 KiB); output arbiter into staging.
- `dma_engine` — ring reader (I2), N-outstanding AXI read master, completion writer, CRC check, error path.
- `staging` — banked SRAM, write arbiter, one read port to an NPU stand-in (a rate-programmable sink, so back-pressure can be studied).
- Parameters: S, B, table type, stream layout, FIFO depths, AXI width, outstanding reads. Fork B build: entropy stage compiled out.

Verification design. The S2 golden model is the reference; cocotb drives descriptors and a Python-side memory model that serves AXI reads with programmable latency and bandwidth (coarse — real DRAM timing is S5's job), and checks I3 output bit-exactly against S2's expected output.

Stimulus tiers: (1) unit — one slot, one block, every S2 vector; (2) constrained-random — random tables, planes, lengths, back-pressure, DMA latency, table changes; (3) real tensors; (4) full-model regression — every block of an 8B archive through a $B = 16$ array (≈ 65 M array-cycles: under an hour with a C++ harness, a few hours through cocotb — overnight is fine). Coverage: Verilator line/toggle; functional points for slot occupancy, FIFO full/empty, every renormalisation case, every table type, descriptor error paths, back-pressure. Assertions in the RTL; the single rANS step proven against its arithmetic definition with `yosys-smtbmc` (boolector/z3 are available) — a small formal target that pays for itself.

Metrics. Symbols/cycle/slot (target ≥ 7.5 of 8 sustained), stall breakdown (input starvation, output back-pressure, table reload), latency per block, ring depth at which throughput saturates, and the slot count for 301 and 452 GB/s at 800 MHz and 1 GHz.

Budget. RTL 4–6 weeks, testbench 2–3 weeks overlapping; sims from minutes (unit) to overnight (full model); RAM small. Fork B build: 2–3 weeks total.

Read. ≥ 7.5 sym/cycle/slot with realistic latency $\rightarrow$ the block-parallel structure holds and D2 goes to `decisions.md`; slot counts feed S4's area multiplication and S5's SRAM sizing. Below that $\rightarrow$ the slot is redesigned before anything is synthesised.

S4 — Synthesis, timing, area, power

Question. What does one slot cost, and does the array fit blueprint D5's $\leq 4 \text{ mm}^2 / \leq 1 \text{ W}$ at 28 nm — as an honest bracket, since no 28 nm PDK is available without an NDA?

Tiers.

1. Yosys `synth generic` $\rightarrow$ cell and gate counts per slot, per table type, per stream layout; `stat -liberty` against Nangate45, ASAP7, sky130 HD, IHP sg13g2 $\rightarrow$ area per slot in each library.
2. OpenSTA on the Yosys netlist + liberty $\rightarrow$ fmax and the critical path per library (post-synthesis, no wires; report with a 30% wire pessimism). Expected critical path: multiply-shift + table lookup + renormalisation select; if it does not close 800 MHz in Nangate45 it will not at 28 nm without pipelining, which changes the sym/cycle/slot picture — back to S3.

3. Optional: OpenROAD place-and-route of one slot and a 4-slot cluster in Nangate45 and ASAP7 for routed area, timing and power (`report_power` with a SAIF from the S3 sim). Hours; several GB of RAM; only if tiers 1–2 leave the D5 answer ambiguous.

SRAM — table copies, FIFOs, staging — from CACTI 7 at 32 nm and 22 nm, which replaces `decoder.py`'s 0.25 mm²/Mbit and `bom.py`'s 1.55 mm²/MB.

Scaling. Report every number per library plus the gate count; project to 28 nm through gate density (`decoder.py`'s 2 M gates/mm² is conservative against 28 nm HPC's ~2.5–3 M) and state the projection as a range. A point value comes from a foundry PDK or C1, not from here.

Budget. 2–3 weeks; ≤ 16 GB RAM with the slot-cluster scoping.

Read. Area per slot × B, power per slot × B, f_{max} — against D5. This is the second half of G1.

S5 — Memory system and pipeline: G3, SRAM and ring sizing

Question. Does 8- or 12-channel DDR5 deliver ≥ 0.80 of peak under the weight-streaming pattern with KV traffic and refresh, and how much staging SRAM and ring depth keep the decoder fed?

Ramulator 2.0. DDR5-4800/5600/6400 device models, x8 devices, 1–2 ranks, 8 and 12 channels × 64-bit; controller FR-FCFS, open vs closed page, address mapping with channel interleave at 256 B / 4 KiB / 64 KiB, 1× refresh; also LPDDR4X-4266 (DRAMsim3 if Ramulator2 lacks it) for the G2-fails branch. Trace frontend, traces generated in Python from the S2 block layout: sequential 64 KiB block reads striped across channels; concurrent KV read/write at 1–3% of weight traffic (per context length); a little activation traffic. Sweep mapping, ranks, page policy, data rate, channel count → achieved GB/s and the efficiency that replaces `BUS_EFFICIENCY = 0.84`.

Pipeline model. A discrete-event model (SimPy in the env, or C++ if it turns out slow): ring (depth D) → DMA (N outstanding, burst size) → memory (Ramulator2-derived latency/bandwidth) → B slots at 8 sym/cycle × clock, with a table-reload penalty on tensor change → staging SRAM (size, banks) → NPU sink at a rate tied to GEMV at bus rate. Token time for 8B (68k blocks) and 70B (600k blocks) — seconds to a minute per token in SimPy — with bus, slot and SRAM utilisation; the minimum SRAM and ring depth for ≥ 0.98 of bus-limited throughput; sensitivity to DMA latency and to the NPU sink rate.

Method check on this PC. Nothing here has 8 channels, so validate the method: model this box's own dual-channel DDR5 in Ramulator2 (DIMM grade is an input — read it from Windows, `wmic memorychip get speed`) and compare with a STREAM-style measurement in WSL. Absolute 8-channel calibration is T2's job.

Budget. Ramulator2 runs minutes each; the DES seconds to minutes; 3–4 weeks of work.

Read. G3 pass at ≥ 0.80. `BUS_EFFICIENCY` becomes a modelled function of (grade, channels, mapping); M6's SRAM and I2's ring depth get numbers.

S6 — Recalibration of models and docs

The simulations exist so that `model/*.py` and `docs/` say what was measured. After S1–S5 and S7:

- `model/system.py` — per-configuration efficiency from S5; a symbol-rate helper keyed on coded bytes per symbol (blueprint D1); predictions for the T1/T2 platforms so the board runs can falsify it.

- `model/decoder.py` — block-parallel structure (slots $\times$ S), per-slot area and power from S4, SRAM from CACTI; the BF16 case kept as one row, not the headline.
- `model/bom.py` — DRAM \$/GB as a scenario (\$2.5 / 5 / 10 / 15), decoder and staging areas from S4, NRE crossover under each scenario.
- `docs/architecture.md` — the lane-count paragraph becomes block-parallel (D2) if S3/S5 confirm; symbol-rate numbers per D1.
- `docs/decisions.md` — D1–D6 in Settled/Ruled out/Reversed as the evidence says; open item 1 closed by E1b, open item 4 by D2.
- `experiments/README.md` and `README.md` — E1's " ~ 175 Gsym/s in ~ 3 mm²" and the "\$460 BOM" become the measured ranges.

One week.

S7 — E2-lite: the software architecture on this PC (G4 at small scale)

Question. On one machine, does one-representation-end-to-end beat the convert-and-materialise path on cold start, swap, and resident footprint — and how far is a software decoder from the bus, on a GPU as well as a CPU?

Setup. Stock llama.cpp (CPU and CUDA builds) is the baseline; models on ext4; page cache dropped between cold-start runs (needs root — or use a fresh copy per run); the storage under WSL2 is a Hyper-V VHDX, so every number is **relative**, same box, same run conditions.

The YFCE-lite path. An lmz page-mapped archive of the model — Q4_K_M GGUF under lmz's block codec today, the S2 format when it lands — plus a loader that demand-pages coded blocks and decodes them into the runtime's weight buffers: CPU via `lmzcore.c` / the S2 C path; GPU via a CUDA rANS block decoder (S7-b, about a week; it doubles as the fast stimulus generator for S3's full-model regression). This is the "coded on flash, coded in transit" half of the invariant, measurable here.

S7-b has since become a project of its own: `vram/`. The CUDA decoder was built in lmz's scratchpad and reached 399–418 GB/s on this card, and the link into that card measures 28.8 GB/s — so the decoder has 14 $\times$ of headroom on the path it feeds, and a half-week experiment turned into a residency engine with four tiers and a placement solver. `vram/MEASURED.md` carries the conditions and `vram/spec/residency.md` restates the four measurements below as what the inference side has to prove.

And then past S7 entirely. The same tier machinery under *training* is a different problem with a much better answer: a training microbatch reuses each weight thousands of times, so at $\sim 2,100$ tokens the offload hides under the compute and the link stops mattering, where decode is short of that by 4000 $\times$ (`decisions.md`, Answered). `vram/` has taken that as its main line — a runtime for full fine-tuning a model too large for its GPU, on CUDA and Metal — and `vram/README.md` is now its charter. S7 remains what it was: the inference measurement, and the baseline the coded tier is scored against.

Measurements, each against stock on identical model and hardware:

1. **Cold start** — flash $\rightarrow$ first token, including the conversion the conventional path needs (HF $\rightarrow$ GGUF convert + quantise) and excluding it.
2. **Model swap** — A $\rightarrow$ B $\rightarrow$ A wall time; **expert swap** on a small MoE that fits 24 GB (OLMoE-1B-7B or Qwen1.5-MoE-A2.7B class).
3. **Resident footprint** — RSS + VRAM for the same model, coded-resident versus materialised.

4. **Sustained tok/s and 1/f** — CPU: coded-in-RAM decoded per token by `lmzcore` (expected far slower — the 1.76 GiB/s wall, already known); GPU: the CUDA block decoder decoding into a VRAM staging buffer per token, reported as GB/s decoded against the 5080's ~806 GB/s achievable. The expected result is that neither software path tracks 1/f. That is the point, and it is what T3-c exists to change.

Half wrong, 2026-08-20. The expectation holds for the CPU, and for the GPU **against VRAM** — 399 GB/s of decode against 776 measured is 0.51×, so a coded tier inside VRAM costs speed and buys capacity. It fails for the GPU **against the link**, where 399 against 28.8 is 14× of headroom and 1/f is tracked exactly. The prediction was made against the wrong bus. `vram/`, and `decisions.md` under Reversed.

Budget. 3–4 weeks including S7-b; runs are minutes.

Read. G4 at PC scale, in relative terms; the README's "materialised four or five times" claim becomes measured; a GPU data point next to the CPU one for "software decode in the bandwidth path".

S8 — Prefill and NPU sizing, and the kernel-model choice, on paper

Three or four days, no simulation. TOPS versus time-to-first-token for 512 / 2k / 8k prompts on 8B, 70B dense, and a 120B-A5B MoE; KV-cache bytes per token versus context length as a fraction of weight traffic; vision and speech encoders at 20 / 60 / 100 TOPS; the die and power cost of each NPU size at 28 nm (`bom.py` floorplan). Output: one table that says, for the AI PC, whether the kernel is a MoE on a 20 TOPS NPU or a dense model on a 60–100 TOPS one — blueprint §3 and D11 — plus the prefill paragraph for `architecture.md`'s "what is deliberately not here".

S9 — The AI OS on this PC: contracts and prototype (G6)

Question. Can the model-as-OS of blueprint §3 be written down as three contracts — event/token bus (I7), interaction (I8: speech first, a glanceable screen, the fixed touch/key set), tool ABI (I9: apps and settings) — and carry an ordinary set of PC tasks **by voice** at these token rates, with no unauthorised tool call under attack, including attacks that arrive as sound? The charter and the first specs are in `os/`. Nothing here needs YFCE silicon; the RTX 5080 stands in for the NPU and this PC's RAM for the DIMMs. What is measured is the *shape* of the AI OS: tokens per interaction, latency classes, state growth, safety — the numbers §3 asserts.

Not bare metal, on purpose. S9 runs as one userspace process under WSL, so it can be built and measured quickly; S10 removes the OS underneath. The process is written the way the runtime will be — one binary, one event loop, static memory plan, no threads it does not own — so that S10 and the YFCE-P runtime are the same code with different drivers.

Build.

- **Runtime prototype** (C, or Rust inside the env): block store over `lmz` page-mapped archives; models through `llama.cpp` as a library (CUDA); persistent KV via its state save/load; a grammar-constrained sampler (GBNF) that is the I7 demultiplexer — UI updates, tool calls, speech, memory writes are the only things the grammar can produce; the widget set and compositor rendering the I8 tree into a WSLg window or a local browser tab at 60 Hz, with caret/scroll/pointer handled by widgets; the I9 tools behind a capability guard with a trusted surface the model cannot draw over; an append-only log; the memory model (`bge-m3`, present) over the log and imported documents.

- **Model set that fits 16 GB:** kernel = Llama-3.1-8B Q4 (4.5 GB), reflex = a 3B-class Q4 (2 GB), streaming ASR (whisper-small/turbo), a wake-word and voice-activity model, a speaker-verification model, a ~0.4B vision encoder, a streaming TTS, bge-m3 — ~9–10 GB plus KV. Voice runs through the PC's own microphone and speakers (or a USB array, T4) via WSLg's audio path; echo cancellation is the prototype's, not the OS's. The kernel/reflex *split* is what is being tested, not the kernel's quality; the 5080's ~800 GB/s makes this 8B "kernel" faster than the real one, so latency classes are reported in tokens as well as seconds and rescaled with `model/system.py`.
- **S9b — the app adapter.** QLoRA on the 8B (10–12 GB VRAM, hours) over synthetic data generated from the I8/I9 specs plus traces of the prototype's own sessions: teaches the UI language and tool grammar, and proves D8 — the adapter is installed as blocks and demand-paged, and switching it is a descriptor change. Its size (tens of MB) is the "app" size.

Measure.

1. **The task set** — 20 ordinary tasks a person does on a PC, **spoken**, half of them with hands busy (write and send a message, look something up and keep it, edit a document, sort photos, set a reminder, do arithmetic over a table, dictate, read a page aloud, import a file from a stick, change the volume / join a network / turn on do-not-disturb, ...). Completed / not, completed hands-free / not, tool calls per task, tokens per step, wall time per step, ASR errors that changed an action, and where a person had to touch or type.
2. **Interaction cost** — the voice pipeline: wake → cue, end of utterance → transcript, → first spoken word (acknowledgement, answer), barge-in → silence, p50/p95, against `os/spec/interaction.md`'s classes; tokens per screen update by kind (heard/doing/need regions, list, table, diff); the diff-vs-full ratio; all on this PC and rescaled to 8 channels with `model/system.py`.
3. **State** — KV growth per hour of use; compaction policy (the model summarising its own context) and its cost; log and index sizes.
4. **Safety** — an injection set (≥ 200 cases: instructions hidden in pages, messages, images, file names, and **audio** — in played media, in a call, in a cloned owner voice, in another language, mid-request) run against every tool; count of unauthorised calls (must be zero, by construction of the guard, not the model's judgement) and of successful *authorised* misuse the trusted surface failed to make visible; speaker-verification false accept/reject under noise and playback.
5. **Boot and swap** — cold start to first reflex token, to kernel; app (adapter) swap latency; a second model set swapped in.

Budget. 5–6 weeks including S9b; runs are minutes; VRAM 16 GB is the constraint (no 70B here — its numbers come from S8 and `system.py`).

Outputs. `os/spec/events.md`, `os/spec/models.md`, and the grammars and signatures that complete `os/spec/interaction.md`, `tools.md` and `trust.md`; the prototype in `os/runtime/`; the adapter in `os/adapters/`; the task set, harness and injection set in `os/eval/`; `os/RESULTS.md` with the five measurements against G6.

Read. G6 pass → the contracts freeze at v0 and D7–D10 go to `decisions.md`; the tokens-per-update figure and latency classes feed S8's NPU/kernel decision. Fail on the task set → the reflex/kernel split or the UI language is redesigned before any hardware conversation about YFCE-P. Fail on safety → the guard, not the model, is redesigned.

S10 — Bare-metal proof on this PC: boots into the model, no OS (optional)

Question. Does the runtime stand up with nothing under it but firmware — and what exactly must firmware provide? This is the "no Windows, no Linux" claim made literal on the machine at hand, and it is what the YFCE-P boot chain and the T2 bare-metal build inherit.

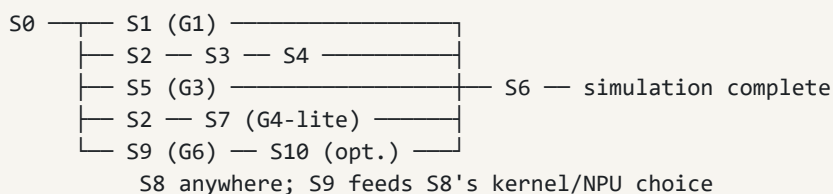
Build. The S9 runtime relinked as a UEFI application: UEFI is the boot ROM here (GOP framebuffer for the compositor, SimpleText/Pointer protocols for input, Block I/O to read archives, SNP for the network tool, timers). CPU-only inference — no GPU driver exists without an OS — with an AVX-512 GEMV over the PC's dual-channel DDR5 (~70 GB/s achievable → 8B Q4 at ~10–15 tok/s, bandwidth-bound like everything else here). Freestanding C, `gcc` that is already installed, gnu-efi vendored and built in-tree; developed and tested in QEMU + OVMF inside the env; run for real from a USB stick. Reads only from the stick; writes nothing to any internal disk; the installed Windows and Linux are untouched and the machine returns to them on reboot.

Measure. Time from power-on to first token; the list of firmware services actually used (this list is the YFCE-P boot-ROM/runtime contract); tok/s versus the STREAM number of the same box (S5's method check); which S9 measurements survive with the OS removed — they should all survive, or the prototype was leaning on Linux somewhere.

Budget. 4–8 weeks. Optional: it changes no gate on its own, but it is the cheapest existence proof of D7 and it is the image T2 boots.

Read. Boots and answers → D7 is demonstrated on commodity hardware; the firmware-service list goes into `blueprint.md` §3 as the runtime's contract with the SoC. Fails on some device → that device's driver joins the staged list in §3, not the list of things a host OS must provide.

Order, dependencies, exit



S1, S5, S7 and S9 do not wait for each other; S3 starts on the common path (DMA, unpack/dequant, staging) the day S2's descriptor record exists and grows the entropy or lattice stage when G1 says which. About seven months serial for one person, four with the parallel branches running; S10 is extra and optional.

Simulation complete means: format vo spec and golden model with vectors; RTL frozen at vo with ≥ 90% line and functional coverage and the metrics above; PPA brackets per library with the 28 nm projection stated as a range; Ramulator2 efficiency curves and DES sizes for SRAM and ring; models and docs recalibrated (S6); the E2-lite report; the AI-OS contracts I7–I9 at vo with the S9 measurements behind them and, if S10 ran, a stick that boots this PC into the model. That package is what `blueprint.md` §8's T1–T4 and C1 consume, and nothing in it required buying anything.